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# Valuation of options under a constant elasticity of variance process and stochastic volatility

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In this paper, we price European call options under a constant elasticity of variance process for the asset price and stochastic volatility. In particular, we derive an analytic approximation formula in the form of asymptotic expansions, which is valid for European options with short times to expiry. Further, we examine the performance of our formula on a market sample of short-tenor crude oil call options (traded on the International Commodities Exchange) and find that the formula provides an excellent fit to market prices.

*Keywords*: Options valuation; Stochastic volatility; Options on oil

## 1. Introduction

The Geometric Brownian Motion (GBM) model can be considered one of the most popular one-factor models for pricing derivatives. The GBM model has been extensively used in the literature (see for example, Brennan and Schwartz 1985, Schwartz 1997). The use of one-factor models has many advantages, mainly for deriving simple and closed pricing formulae for derivatives. However, various two-factor models have been introduced in the literature to achieve more appropriate derivatives pricing formulae. The second factor for pricing commodity options is usually taken to be either the convenience yield or long-run price (see, for example, Ribeiro and Hodges 2004, Schwartz 1997, Pilipovic 1997).

One of the most recognised works on pricing options was done by Black and Scholes (1973), who derived the well-known closed-form formula for pricing European options. However, the Black–Scholes formula can lead to serious mispricing, as suggested by some empirical studies (see, for example, Duan 1999). Consequently, various stochastic volatility models have been developed (see, for example, Hull and White 1987, Scott 1987, Wiggins 1987). One of the most popular works on stochastic volatility models was conducted by Heston (1997) who derived a closed-form formula for the pricing of European options when the spot price of the underlying asset is correlated with its variance (volatility squared) that is assumed to follow a square root mean-reverting model, namely, $\mathrm{d}v = k(a-v)\,\mathrm{d}t + c\sqrt{v}\,\mathrm{d}Z$. Some studies have investigated the performance of various other variance models. For example, Chacko and Viceira (2003) compared the performance of stochastic volatility models of the form $\mathrm{d}v = (a+bv)\mathrm{d}t + cv^{\gamma}\,\mathrm{d}Z$, and concluded that these models with $1 \le \gamma \le 2$ performed better than other models. Similarly, Chrisoffersen *et al.* (2010) compared stochastic volatility models of the form $\mathrm{d}v = k_1 v^{k_2}(a-v)\,\mathrm{d}t + cv^{k_3}\,\mathrm{d}Z$ where $k_2 = \{1, 0\}$ and $k_3 = \{0.5, 1, 1.5\}$ and found that two models (namely, with $k_2 = 1, k_3 = 1.5$ and $k_2 = 0, k_3 = 1$) performed better in fitting various samples of S&P500 index returns. Goard and Mazur (2013) empirically show that the 3/2-model

$$\mathrm{d}S = \mu S\,\mathrm{d}t + \sqrt{v}S\,\mathrm{d}W_1 \tag{1a}$$

$$\mathrm{d}v = (av - bv^2)\,\mathrm{d}t + bv^{\frac{3}{2}}\,\mathrm{d}W_2 \tag{1b}$$

outperforms all other models considered in their tests, including the Heston model, in fitting the VIX time series.

The aim of this paper is to price European call options on an underlying with price $S$ that follows the volatility risk-neutral model, namely

$$\mathrm{d}S = rS\,\mathrm{d}t + \sqrt{v}S^{\gamma_1}\,\mathrm{d}\tilde{W}_1 \tag{2a}$$

$$\mathrm{d}v = \eta v(a-v)\,\mathrm{d}t + bv^{\gamma_2}\,\mathrm{d}\tilde{W}_2 \tag{2b}$$

for some constants $r$, $\gamma_1$, $\eta$, $a$, $b$, $\gamma_2$ and $-1 \le \rho \le 1$ and where $\mathbb{E}(\mathrm{d}\tilde{W}_1\mathrm{d}\tilde{W}_2) = \rho\mathrm{d}t$. In our proposed model (2a), (2b), $\mathrm{d}\tilde{W}_1$ and $\mathrm{d}\tilde{W}_2$ represent increments in standard Wiener processes under an equivalent risk-neutral probability measure

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$\mathbb{Q}$. Furthermore, $r$ represents the constant risk-free rate and $v$ is the variance. Equation (2b) assumes that the variance reverts to a constant $a$ with the rate of reversion determined by $\eta v$, and $b$ is the volatility of volatility. It should be noted that the proposed model (2a), (2b) allows the instantaneous standard deviation of the underlying asset price and variance to be proportional to any power of asset price and variance.

The derived pricing formula is in the form of an asymptotic expansion. The asymptotic expansion method is used extensively in the literature for the valuation of options contracts (an outline of this method can be found in Howison (2005)). Recently, AbaOud and Goard (2016) derived an analytic approximation formula for pricing European crack spread call options under both a univariate and explicit constant elasticity of variance (CEV) model and empirically showed that their formulae outperform other models considered in their tests in capturing market prices.

The remainder of this paper is organised into four sections. In section 2, an analytic approximation formula for the pricing of European call option contracts under the model (2a,b) is derived and is shown to be accurate for pricing short-tenor options in section 3. The results of an empirical test that measures the performance of our formula in its ability to capture market prices is presented in section 4. In section 5, we present our conclusions.

## 2. Deriving the valuation formula

Given that the underlying asset price follows (2a,b), then by letting $\tau = T - t$, the price of a European call option, $V(S, v, \tau)$, with strike price $K$ and expiry $T$, is governed by the following partial differential equation (PDE):

$$\frac{\partial V}{\partial \tau} = \frac{vS^{2\gamma_1}}{2}\frac{\partial^2 V}{\partial S^2} + \frac{b^2 v^{2\gamma_2}}{2}\frac{\partial^2 V}{\partial v^2} + \rho b S^{\gamma_1} v^{\gamma_2+\frac{1}{2}}\frac{\partial^2 V}{\partial S \partial v} + rS\frac{\partial V}{\partial S} + \eta v(a - v)\frac{\partial V}{\partial v} - rV \tag{3a}$$

subject to

$$V(S, v, 0) = \max(S - K, 0) \tag{3b}$$

$$\lim_{S\to 0} V(S, v, \tau) = 0 \tag{3c}$$

$$\lim_{S\to \infty} V(S, v, \tau) \sim S \tag{3d}$$

$$\lim_{v\to 0} V(S, v, \tau) = \max(S - Ke^{-r\tau}, 0) \tag{3e}$$

$$\lim_{v\to \infty} V(S, v, \tau) = S \tag{3f}$$

The boundary conditions for $S$ (i.e. (3c), (3d)) are known as the standard conditions for European call options (see, for example, Hull 2012). The boundary condition (3e) is the riskless growth for the underlying asset $S$, while the boundary condition (3f) is derived from the Black–Scholes formula for the call option by taking $\sqrt{v} \to \infty$ (see Zhu and Chen 2011).

In the following theorem 2.1, we provide a simple approximate solution to (3a)–(3f) valid for short times to expiry which only involves simple mathematical functions. The proof of theorem 2.1 is based on the perturbation method, which requires to have (or to introduce) a small parameter $0 < \varepsilon \ll 1$ and assumes that the solution of the PDE can be written as a series in $\varepsilon$. Options with short tenor are the most options traded in the market, hence we assume that the time to expiry is short i.e.$\tau = \varepsilon\hat{\tau}$.

THEOREM 2.1 *Given that the underlying asset follows the risk-neutral process* (2a), (2b)*, then an approximate solution for a European call option with exercise price K and expiry T valid for short times to expiry is given by*

$$\begin{aligned} V(S, v, \tau) = \Bigg( & K\sqrt{\frac{\theta_1 \tau}{\pi}} + \frac{(S-K)\sqrt{\tau}(2v\theta_2 - \theta_3)}{8v\sqrt{\pi\theta_1}} \\ & + \frac{3(S-K)^4 A_1}{768K^3\sqrt{\pi\theta_1\tau}} + \frac{(S-K)^2\sqrt{\tau}A_2}{48K\sqrt{\pi\theta_1}\theta_1 v^4} \\ & - \frac{K\tau^{\frac{3}{2}}A_3}{24\sqrt{\pi\theta_1}v^4}\Bigg) e^{-\frac{(S-K)^2}{4K^2\theta_1\tau}} \\ & + \frac{S - K(1 - r\tau)}{2}\,\mathrm{erfc}\left(-\frac{S-K}{2K\sqrt{\theta_1\tau}}\right) \end{aligned} \tag{4}$$

where
$$\begin{aligned} \theta_1 &= \frac{vK^{2\gamma_1-2}}{2} \\ \theta_2 &= \gamma_1 v K^{2\gamma_1-2} \\ \theta_3 &= \rho b v^{\gamma_2+\frac{1}{2}} K^{\gamma_1-1} \\ A_1 &= \frac{(2\theta_2 v - \theta_3)^2}{v^2{\theta_1}^2} \\ A_2 &= \theta_1 b^2 v^{2\gamma_2+2} + 4\theta_1\gamma_1 v^4(-3\gamma_1\theta_1 - 3r + 2\theta_1) \\ &\quad + 3v^3\theta_3 r + \theta_3^2 v^2\left(\gamma_2 - \frac{7}{4}\right) \\ A_3 &= 6\theta_1 v^5\eta + (-6a\eta\theta_1 - 6r^2 + 12r\theta_1 \\ &\quad + 2\gamma_1\theta_1^2(3\gamma_1 - 2))v^4 + 3\theta_3 v^3\left(r - \frac{\theta_2}{2}\right) \\ &\quad + \theta_3^2\left(\gamma_2 - \frac{5}{8}\right)v^2 + b^2\theta_1 v^{2\gamma_2+2} \end{aligned}$$

*Proof* For a short time to expiry $\tau$, we let $\tau = \varepsilon\hat{\tau}$ where $0 < \varepsilon \ll 1$. Then equation (3a) becomes

$$\frac{\partial V}{\partial \hat{\tau}} = \varepsilon\left(\frac{vS^{2\gamma_1}}{2}\frac{\partial^2 V}{\partial S^2} + \frac{b^2 v^{2\gamma_2}}{2}\frac{\partial^2 V}{\partial v^2} + \rho b S^{\gamma_1} v^{\gamma_2+\frac{1}{2}}\frac{\partial^2 V}{\partial S\partial v} + rS\frac{\partial U}{\partial S} + \eta v(a-v)\frac{\partial V}{\partial v} - rV\right) \tag{5}$$

We assume that the solution can be written as a series in $\varepsilon$, i.e.

$$V(S, v, \hat{\tau}) = \sum_{i=0}^{\infty} \varepsilon^i V_i(S, v, \hat{\tau}). \tag{6}$$

Substituting (6) into (5), and equating the coefficients of $\varepsilon^m$ (where $m = 0, 1, 2$) on both sides of the resulting equation, we get

$$\varepsilon^0 : \frac{\partial V_0}{\partial \grave{\tau}} = 0 \tag{7a}$$

$$\varepsilon^1 : \frac{\partial V_1}{\partial \grave{\tau}} = \frac{vS^{2\gamma_1}}{2}\frac{\partial^2 V_0}{\partial S^2} + \frac{b^2 v^{2\gamma_2}}{2}\frac{\partial^2 V_0}{\partial v^2} + \rho b S^{\gamma_1} v^{\gamma_2+\frac{1}{2}} \frac{\partial^2 V_0}{\partial S \partial v} + rS\frac{\partial V_0}{\partial S} + \eta v(a-v)\frac{\partial V_0}{\partial v} - rV_0 \tag{7b}$$

$$\varepsilon^2 : \frac{\partial V_2}{\partial \grave{\tau}} = \frac{vS^{2\gamma_1}}{2}\frac{\partial^2 V_1}{\partial S^2} + \frac{b^2 v^{2\gamma_2}}{2}\frac{\partial^2 V_1}{\partial v^2} + \rho b S^{\gamma_1} v^{\gamma_2+\frac{1}{2}} \frac{\partial^2 V_1}{\partial S \partial v} + rS\frac{\partial V_0}{\partial S} + \eta v(a-v)\frac{\partial V_1}{\partial v} - rV_1 \tag{7c}$$

Equations (7a)–(7c) need to be solved subject to

$$V_0(S, v, 0) = \max(S-K, 0) \tag{8a}$$
$$V_1(S, v, 0) = 0 \tag{8b}$$
$$V_2(S, v, 0) = 0 \tag{8c}$$

Solving (7a)–(7c) subject to (8a)–(8c) we get the first three terms of our series solution as:

$$V_0(S, v, \grave{\tau}) + \varepsilon V_1(S, v, \grave{\tau}) + \varepsilon^2 V_2(S, v, \grave{\tau}) = \begin{cases} S - K + \varepsilon K r\grave{\tau} - \varepsilon^2 \dfrac{Kr^2\grave{\tau}^2}{2} & \text{if } S-K \gg \varepsilon K, \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Equation (9) suggests a corner layer at $S = K$. Thus, solution (9) is invalid in this region and termed as an 'outer' solution. In order to analyse the solution in the inner region, we introduce a stretching variable

$$x = \frac{S-K}{\sqrt{\varepsilon}K} \tag{10a}$$

and rescale

$$V(S, v, \grave{\tau}) = K\sqrt{\varepsilon}\, W(S, v, \grave{\tau}). \tag{10b}$$

Equation (5) then becomes

$$\frac{\partial W}{\partial \grave{\tau}} = \frac{K^{2\gamma_1-2} v(1+\sqrt{\varepsilon}x)^{2\gamma_1}}{2}\frac{\partial^2 W}{\partial x^2} + \varepsilon\frac{b^2 v^{\gamma_2}}{2}\frac{\partial^2 W}{\partial v^2} + \sqrt{\varepsilon}\rho b(1+\sqrt{\varepsilon}x)^{\gamma_1} v^{\gamma_2+\frac{1}{2}}\frac{\partial^2 W}{\partial x \partial v} + \sqrt{\varepsilon} r(1+\sqrt{\varepsilon}x)\frac{\partial W}{\partial x} + \varepsilon\eta v(a-v)\frac{\partial W}{\partial v} - \varepsilon r W \tag{11}$$

which needs to be solved subject to the conditions $W(x, v, 0) = \max(x, 0)$, $W(x, v, \grave{\tau}) \to x + \sqrt{\varepsilon} r\grave{\tau} + o(\varepsilon^{\frac{3}{2}})$ as $x \to \infty$ and $W(x, v, \grave{\tau}) \to 0$ as $x \to -\infty$.

Now we expand $W(x, v, \grave{\tau})$ in terms of $\sqrt{\varepsilon}$, i.e.

$$W(x, v, \grave{\tau}) = \sum_{i=0}^{\infty} \varepsilon^{\frac{i}{2}} W_i(x, v, \grave{\tau}) \tag{12}$$

Substituting (12) into (11), and equating the coefficients of $\varepsilon^m$ (where $m = 0, 1, 2$) on both sides of the resulting equation, we get

$$(\sqrt{\varepsilon})^0 : \frac{\partial W_0}{\partial \grave{\tau}} = \theta_1 \frac{\partial^2 W_0}{\partial x^2} \tag{13a}$$

$$(\sqrt{\varepsilon})^1 : \frac{\partial W_1}{\partial \grave{\tau}} = \theta_1 \frac{\partial^2 W_1}{\partial x^2} + \theta_2 x\frac{\partial W_0}{\partial x^2} + r\frac{\partial W_0}{\partial x} + \theta_3\frac{\partial^2 W_1}{\partial x \partial v} \tag{13b}$$

$$(\sqrt{\varepsilon})^2 : \frac{\partial W_2}{\partial \grave{\tau}} = \theta_1 \frac{\partial^2 W_2}{\partial x^2} + q(x, v, \grave{\tau}) \tag{13c}$$

$$\text{as } q(x, v, \grave{\tau}) = r\left(x\frac{\partial W_0}{\partial x} + \frac{\partial W_1}{\partial x} - W_0\right) + \theta_3\left(\gamma_1 x\frac{\partial^2 W_0}{\partial x \partial v} + \frac{\partial^2 W_1}{\partial x \partial v}\right) + \eta v(a-v)\frac{\partial W_0}{\partial x} + \frac{b^2 v^{2\gamma_2}}{2}\frac{\partial^2 W_0}{\partial v^2} + 2\theta_1\gamma_1 x\left(\frac{\partial^2 W_1}{\partial x^2} + \frac{x}{2}\frac{\partial^2 W_0}{\partial x^2}\right)$$

Equation (13a) needs to be solved subject to $W_0(x, v, 0) = max(x, 0)$, $W_0(x, v, \grave{\tau}) \to x$ as $x \to \infty$ and $W_0(x, v, \grave{\tau}) \to 0$ as $x \to -\infty$. The solution to (13a) is easily found to be†

$$W_0(x, v, \grave{\tau}) = \sqrt{\frac{\theta_1\grave{\tau}}{\pi}}\, e^{-\frac{x^2}{4\theta_1\grave{\tau}}} + \frac{x}{2}\operatorname{erfc}\left(-\frac{x}{2\sqrt{\theta_1\grave{\tau}}}\right), \tag{14}$$

The conditions for equation (13b) are $W_1(x, v, 0) = 0$, $W_1(x, v, \grave{\tau}) \to r\grave{\tau}$ as $x \to \infty$ and $W_1(x, v, \grave{\tau}) \to 0$ as $x \to -\infty$. The particular solution‡ of (13b) satisfying the given conditions is found as

$$W_1(x, v, \grave{\tau}) = \frac{x\sqrt{\grave{\tau}}(2v\theta_2 - \theta_3)}{8v\sqrt{\pi\theta_1}}\, e^{-\frac{x^2}{4\theta_1\grave{\tau}}} + \frac{r\grave{\tau}}{2}\operatorname{erfc}\left(-\frac{x}{2\sqrt{\theta_1\grave{\tau}}}\right), \tag{15}$$

Equation (13c) needs to be solved subject to the conditions $W_2(x, v, 0) = 0$ and $W_2(x, v, \grave{\tau}) \to 0$ as $x \to \pm\infty$. The solution of (13c) is given by

$$W_2(x, v, \grave{\tau}) = \int_0^{\grave{\tau}}\int_{-\infty}^{\infty} \frac{1}{2\sqrt{\pi\theta_1(\grave{\tau}-\omega)}}\, q(\zeta, v, \omega) e^{-\frac{(x-\zeta)^2}{4\theta_1(\grave{\tau}-\omega)}}\, \mathrm{d}\zeta\, \mathrm{d}\omega$$

† By letting $W_0(x, \grave{\tau}) = \sqrt{\grave{\tau}}\phi(z)$, where $z = \frac{x}{\sqrt{\grave{\tau}}}$, this substitution leads to an ODE in $\phi(z)$.

‡ We used the following two results: (i) if $u_y = \frac{1}{2}u_{xx}$; and $v_y = \frac{1}{2}v_{xx} + u$, then $v = yu$ is a particular solution and (ii) if $u_y = \frac{1}{2}u_{xx}$ and $v_y = \frac{1}{2}v_{xx} + xu$, then $v = xyu + \frac{y^2}{2}u_x$ is a particular solution.

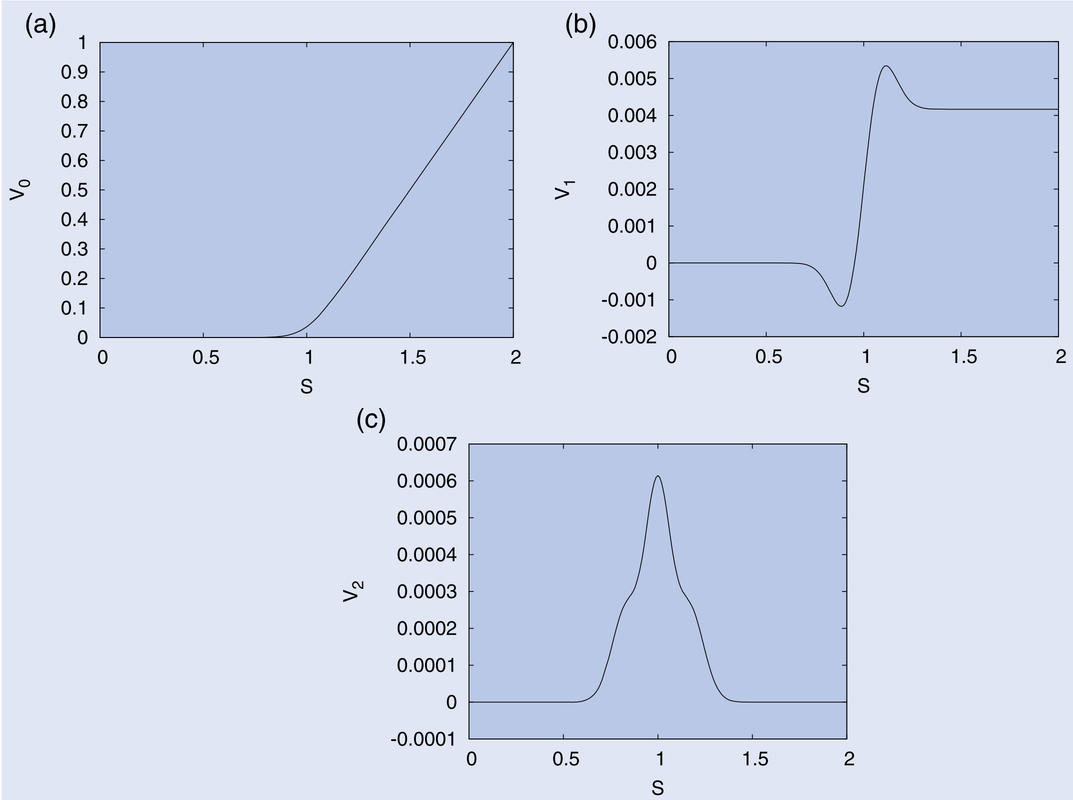


Figure 1. Comparison of the first three terms solutions of (4) with $\tau = \frac{1}{12}$, $v = 0.1$, $a = 0.2$ $r = 0.05$, $\gamma_1 = 1$, $\gamma_2 = 1.5$, $K = 1$, $\eta = 10$, $b = 2$, and $\rho = -0.70$ ((a) $V_0$, (b) $V_1$ and (c) $V_2$).

$$= \frac{e^{-\frac{x^2}{4\theta_1 \grave{\tau}}}}{24\sqrt{\pi\theta_1}} \left( \frac{3x^4 A_1}{32\sqrt{\grave{\tau}}} + \frac{x^2\sqrt{\grave{\tau}}A_2}{2\theta_1 v^4} - \frac{\grave{\tau}^{\frac{3}{2}} A_3}{v^4} \right) \tag{16}$$

Solutions (14), (15) and (16) lead to the three-term inner expansion, namely,

$$W(x, v, \grave{\tau}) = W_0(x, v, \grave{\tau}) + \sqrt{\varepsilon} W_1(x, v, \grave{\tau}) + \varepsilon W_2(x, v, \grave{\tau}) \tag{17}$$

Now, we combine the outer and inner expansions, i.e. (9) and (17), and then subtract the common part (in our case the common part coincides with the outer solution) to get the uniform expansion. Hence, we get the price of a call option contract by undoing the change of variables, i.e. (10) in (17) to get (4). ■

Figure 1 shows a comparison of the first three terms solutions of (4). From this figure with the given parameter values, it can be seen that the values of consecutive terms are getting smaller and smaller. This is an indication that the higher-order terms (i.e. after the second-order approximation) can be neglected, and hence, the second-order approximation can be considered a sufficient approximation for (3a)–(3f).

## 3. The accuracy of the analytic approximation formula

In this section, we provide numerical examples to measure the accuracy of our new analytic approximation formula (4). We considered three values of time to expiry $T - t = \{\frac{1}{12}, \frac{2}{12}, \frac{3}{12}\}$ and three pairs of diffusion powers $(\gamma_1, \gamma_2) = \{(0.75, 1), (1, 1), (1, 1.5)\}$. For $(\gamma_1, \gamma_2) = (1, 1.5)$, results obtained from formula (4) are compared with corresponding exact solutions† of (3a), which were obtained by Goard (2014). For $(\gamma_1, \gamma_2) = \{(0.75, 1), (1, 1)\}$, these is no known exact solution to (3a). Hence, results obtained from formula (4) are compared with the numerical solutions‡ of (3a), which are assumed as a proxy for the exact solution. For each value of $T - t$ and $(\gamma_1, \gamma_2)$, we chose typical values $v = 0.1, K = 1, r = 0.05, \eta = 10, b = 2, a = 0.2, \rho = -0.70$ and three values of the underlying asset price ($S$). The signed percentage error (SPE):

$$\text{SPE} = \frac{C_{\text{exact}} - C_{\text{approximate}}}{C_{\text{exact}}} \times 100\%$$

and absolute error (AE) for each couple $(S, (\gamma_1, \gamma_2))$ were calculated. Table 1 lists the results, from which we note that:

The analytic approximation formula (4) generally slightly overprices option contracts compared with the exact solution. For all the chosen $(S, (\gamma_1, \gamma_2))$ with $T - t = \frac{1}{12}$, the SPE values lie within a small range, namely, $(-0.35\%, 0.06\%)$. As time to expiry increases, the SPE values increase (as expected), but for all the chosen $(S, (\gamma_1, \gamma_2))$ with $T - t = \frac{3}{12}$, the SPE values remain small and lie within

† Goard 2014 found an exact solution for pricing European put options; we used the put-call parity to obtain the price of European call options.
‡ The central implicit finite-difference method (with $\Delta\tau = 1 \times 10^{-6}$, $\Delta S = 0.01$ and $\Delta v = 0.025$) is used to obtain numerical solutions of (3a).

Table 1. Signed percentage error (SPE) and absolute error (AE) of analytic approximation formulae (4) with $v = 0.1, K = 1, r = 0.05, \eta = 10, b = 2, a = 0.2, \rho = -0.70$.

| $(\gamma_1, \gamma_2)$ | | (0.75, 1) | | (1, 1) | | (1, 1.5) | |
|---|---|---|---|---|---|---|---|
| $T-t$ | $S$ | AE($\times 10^{-3}$) | SPE (%) | AE($\times 10^{-3}$) | SPE (%) | AE($\times 10^{-3}$) | SPE (%) |
| $\frac{3}{12}$ | | | | | | | |
| | 0.95 | 0.54 | − 1.31 | 0.56 | − 1.37 | 0.14 | − 0.30 |
| | 1.05 | 1.66 | − 1.63 | 1.59 | − 1.56 | 0.58 | − 0.56 |
| | 1.15 | 2.54 | − 1.40 | 2.75 | − 1.51 | 0.54 | − 0.30 |
| Average | | 1.58 | | 1.64 | | 0.42 | |
| $\frac{2}{12}$ | | | | | | | |
| | 0.95 | 0.14 | − 0.47 | 0.17 | − 0.60 | 0.02 | − 0.05 |
| | 1.05 | 0.63 | − 0.71 | 0.62 | − 0.70 | 0.24 | − 0.27 |
| | 1.15 | 0.94 | − 0.55 | 1.15 | − 0.67 | 0.16 | − 0.10 |
| Average | | 0.57 | | 0.65 | | 0.14 | |
| $\frac{1}{12}$ | | | | | | | |
| | 0.95 | 0.02 | − 0.16 | 0.05 | − 0.35 | 0.01 | 0.06 |
| | 1.05 | 0.14 | − 0.20 | 0.16 | − 0.22 | 0.05 | − 0.08 |
| | 1.15 | 0.11 | − 0.07 | 0.23 | − 0.15 | 0.00 | 0.00 |
| Average | | 0.09 | | 0.15 | | 0.02 | |

$(-1.63\%, -0.30\%)$. The average absolute errors are less than $0.09 \times 10^{-3}$, $0.46 \times 10^{-3}$ and $1.21 \times 10^{-3}$ with $T - t = \frac{1}{12}$, $\frac{2}{12}$ and $\frac{3}{12}$, respectively. This suggests that the analytic approximation formula (4) provides an excellent approximation to the exact solution.

## 4. Calibration to market data

The options data used in our calibration was selected from the International Commodities Exchange (ICE) and consisted of three groups of observations of call option† closing prices in the year 2019, with the underlying asset being Brent crude oil futures contracts. The first, second and third groups have times to expiry of 30, 60 and 90 days, respectively. Each group has a known underlying price ($S$) (which is the price of the futures contract on the selected trading date), an estimated variance ($v$) (which is assumed to be the variance of the futures prices in the previous seven working days prior to the selected trading date) and various strike prices. For each group, we estimate the other parameters‡ $(\gamma_1, \eta, a, b, \gamma_2, \rho)$, which are chosen to provide the 'best' fit to our options data by minimising the sum of squares of errors (SSE). Table 2 lists the results of the estimated parameters, SSE and adjusted root mean square error (ARMSE), which takes into account the number of parameters and observations. In particular, we note that

- Our formula (4) provides an excellent fit for all three groups (see figure 2). The ARMSE values are \$0.0133, \$0.0167 and \$0.0205 (per contract) for the first, second and third group, respectively. The ARMSE value (as expected) slightly increases with time to expiry.
- The estimated $\gamma_1$ values ($\{0.8530, 0.8820, 0.8421\}$) are close to the empirical results found in AbaOud and Goard (2015), namely, the one factor $dF = \sigma F^{\frac{3}{4}} dW$ which performed best in fitting market option (on oil futures) prices compared with the GBM and Schwartz models.
- The estimated $\gamma_2$ values ($\{1.5056, 1.8552\ 1.7952\}$) are in agreement with the empirical results found in Chacko and Viceira (2003), namely, the stochastic volatility models with $1 \le \gamma \le 2$ performed better than other models tested of the form $dv = (a + bv)\, dt + cv^{\gamma} dW$.
- The estimated $\rho$ values are negative. These values are expected, as asset price shocks are more likely to be negatively correlated with volatility shocks (see Goard 2014).

For each group, we used the estimated parameters to compute model prices for each strike price. Then the corresponding signed percentage error (SPE), SPE $= \frac{(V_{\text{market}} - V_{\text{model}})}{V_{\text{market}}} \times 100\%$ , is calculated for each option contract. Given that the moneyness function ($M$) is defined as $M = \ln(\frac{S}{K})$, the average signed percentage errors were calculated for the following three groups of moneyness: in-the-money options (ITM) with $M > 0.05$, at-the-money options (ATM) with $M \in [-0.05, 0.05]$ and out-of-the-money options (OTM) with $M < -0.05$. These averages are also listed in table 2. Comparison of average signed percentage errors indicates that our model overprices ITM (also for OTM options with one month to expiry) compared with market prices. This overpricing could be interpreted as the market demand on these options being significantly high. For ATM and OTM§ options, model prices are lower than market prices, suggesting that the market demand on these options could be low.

† These options are quoted as American style. However, the early exercise premiums are assumed to be negligible for three reasons: futures do not pay dividends, times to expiry are chosen to be small and the interest rate was low.
‡ Under the assumption that the risk-neutral futures prices follow (2a) with zero drift, i.e. $dS = \sqrt{v} S^{\gamma_1} d\tilde{W}_1$.

§ It must be noted that both market and model prices for some observations are less than 0.05 yielding high average percentage errors for these options.

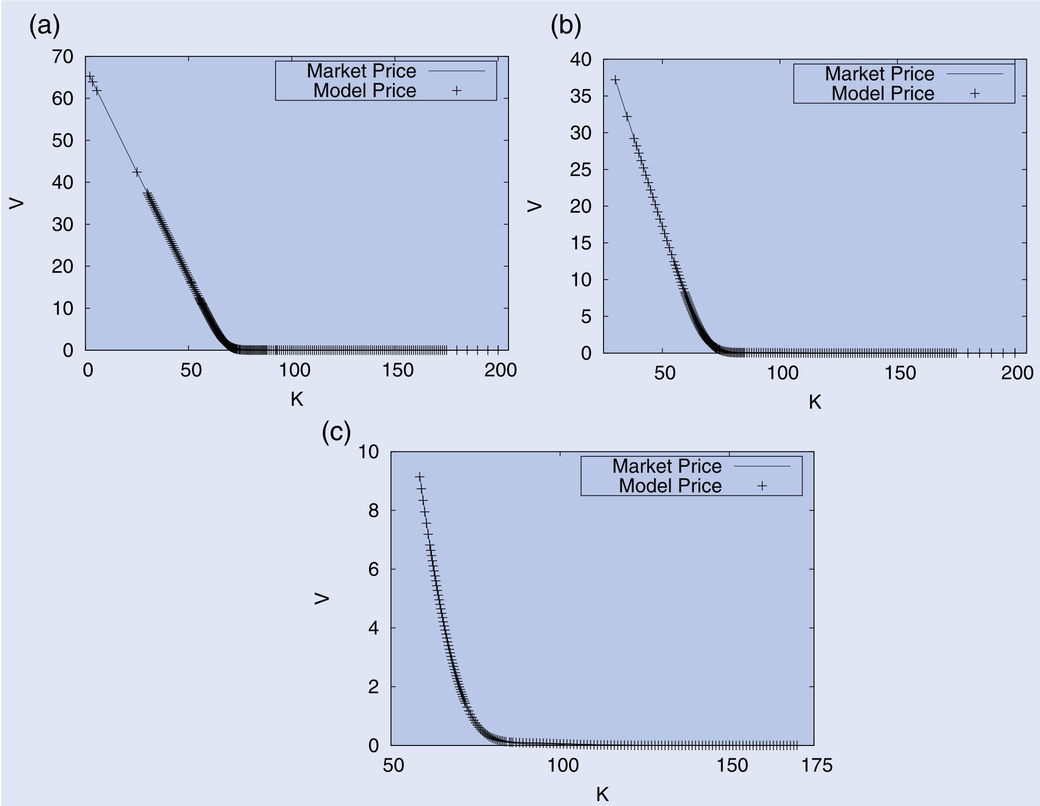


Figure 2. Comparison of options prices ($V$) using formula (4) with market data. The estimated parameters are listed in table 2 ((a) for Group 1 with $\tau = \frac{1}{12}$, (b) for Group 2 with $\tau = \frac{2}{12}$ and (c) for Group 3 with $\tau = \frac{3}{12}$).

Table 2. Estimated parameters, SSE, ARMSE and average SPE of option pricing using formula (4).

| | | Group 1 | Group 2 | Group 3 |
|---|---|---|---|---|
| Given values | $S$ | 67.43 | 67.21 | 66.62 |
| | $v$ | 0.2427 | 0.2496 | 0.2333 |
| Estimated parameters | $\gamma_1$ | 0.8820 | 0.8530 | 0.8421 |
| | $\eta$ | 31.7911 | 32.0173 | 31.9738 |
| | $a$ | 0.1350 | 0.2295 | 0.2959 |
| | $b$ | 7.9994 | 7.8884 | 3.0570 |
| | $\gamma_2$ | 1.5056 | 1.8552 | 1.7952 |
| | $\rho$ | $-0.3003$ | $-0.3793$ | $-0.9998$ |
| Error | # of obs. | 255 | 204 | 160 |
| | *SSE* | 0.0442 | 0.0555 | 0.0648 |
| | *ARMSE* | 0.0133 | 0.0167 | 0.0205 |
| Average SPE | ITM | $-0.05$ | $-0.09$ | $-0.38$ |
| | $M \in$ | $(0.05, 0.99]$ | $(0.05, 0.81]$ | $(0.05, 0.13]$ |
| | ATM | 0.45 | 0.34 | 0.40 |
| | $M \in$ | $[-0.05, 0.05]$ | $[-0.05, 0.05]$ | $[-0.05, 0.05]$ |
| | OTM | $-5.33$ | 5.76 | 8.27 |
| | $M \in$ | $[-0.18, -0.05)$ | $[-0.25, -0.05)$ | $[-0.37, -0.05)$ |

## 5. Conclusion

Under the volatility risk-neutral model (2a), (2b), we found an analytic approximate solution for European call options with short tenor. Numerical examples suggest that the analytic approximation formula (4) provides an excellent approximation to the exact solution for options with times to expiry up to three months. We examined our formula in capturing market prices of options on oil futures and the results provided an excellent fit to market prices. Furthermore, the estimated values of the diffusion powers (i.e. $\gamma_1$ and $\gamma_2$) are consistent with other empirical results in the literature. It could be anticipated that the analytic approximation formula (4) under model (2a,b) provides useful guides to traders.

## Data availability

The data used in this research is available on the International Commodities Exchange (ICE) via https://www.ice.if5.com.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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